

CSE206 (Digital Logic Design Sessional)

Expt. No. 9

Name of the Expt.

Registers

7

<u>Group No.</u>	1
<u>Writer's Roll</u>	1705094
<u>Section</u>	B
<u>Department</u>	CSE
<u>Other Group Members</u>	1705091 1705092 1705093 1705095
<u>Date of Performace:</u>	4/9/19
<u>Date of Submission:</u>	11/9/19

Problem specification

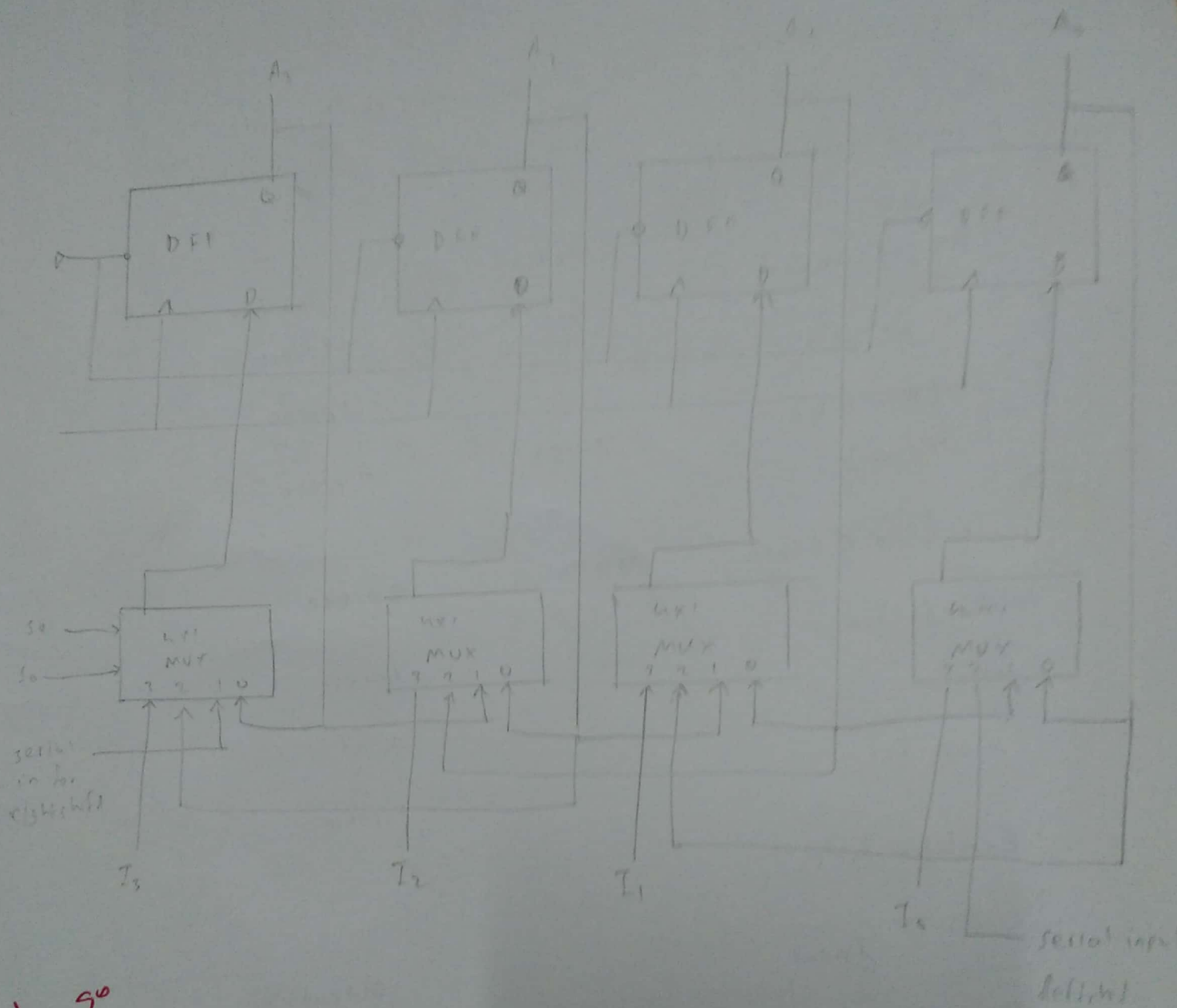
Design and implement a 4-bit universal shift register

Required instruments

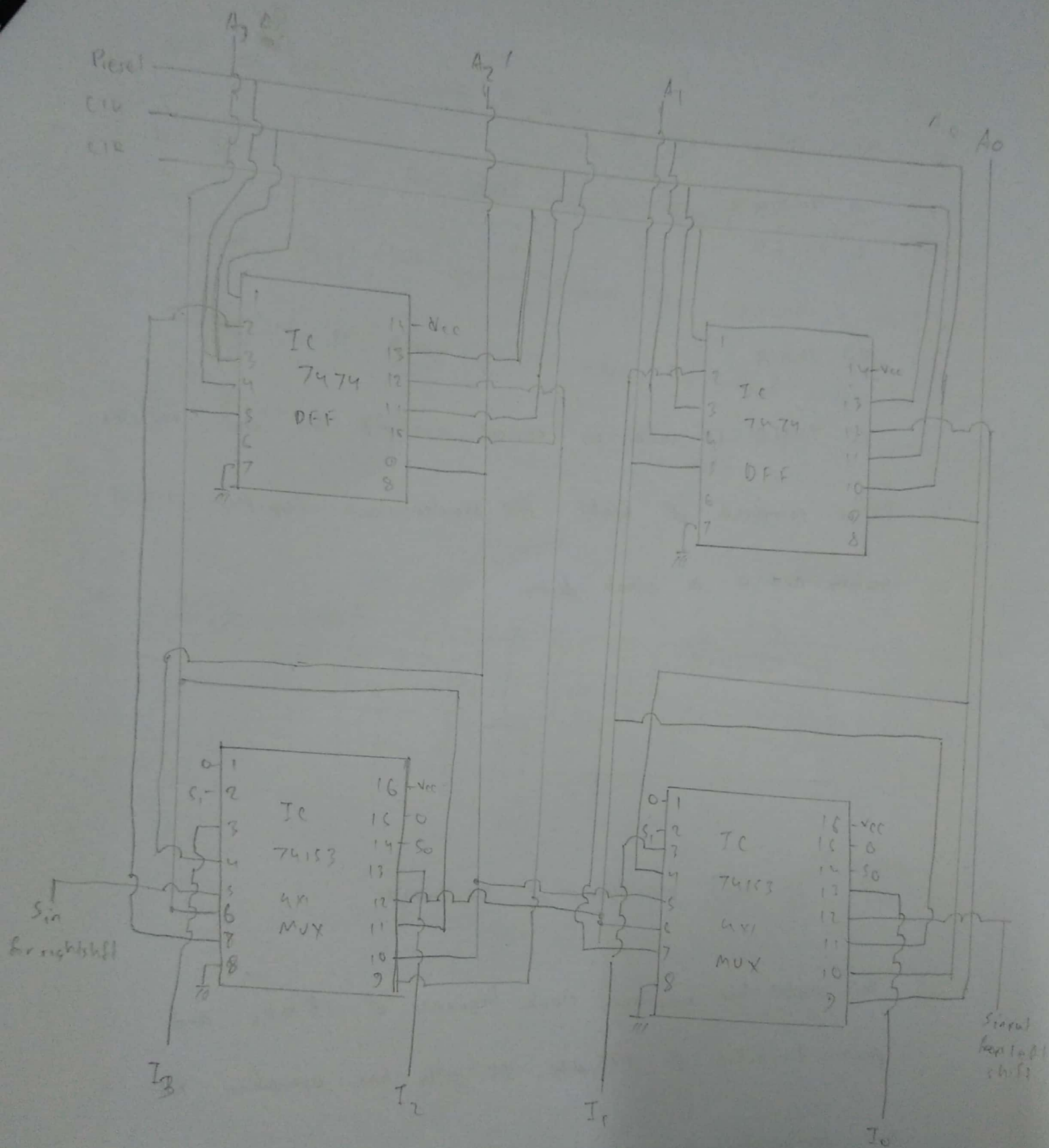
Name	Model	Quantity
Trainer board		1 piece
IC- Extractor		1 piece
IC (Dual 4 to 1 multiplexer)	74153	2 piece
IC (Dual D FlipFlop)	7474	2 piece
Hand book		1 piece
Wire		a lot
Magnifier		1 piece

Excitation table :

Input					Output				
clear	clock	s_0	s_1	Internal signal $D_i (i=0,1,2,3)$	A_3	A_2	A_1	A_0	Mode
0	x	x	x	x	0	0	0	0	Asynchronous
1	x	0	0	x	A_3	A_2	A_1	A_0	data hold
1	↑	0	1	A_{i+1}	A_2	A_1	A_0	I_2	shift left
1	↑	1	0	A_{i-1}	I_2	A_3	A_2	A_1	shift right
1	↑	1	1	A_i	I_3	I_2	I_1	I_0	Parallel load



why so
right?



Write short note on

a. 7491A

b. 74164

IC-7491A :

IC 7491A is serial-in, serial out 8 bit shift register.

It is composed of eight R-S master-slave flip-flops, input gates and a clock driver.

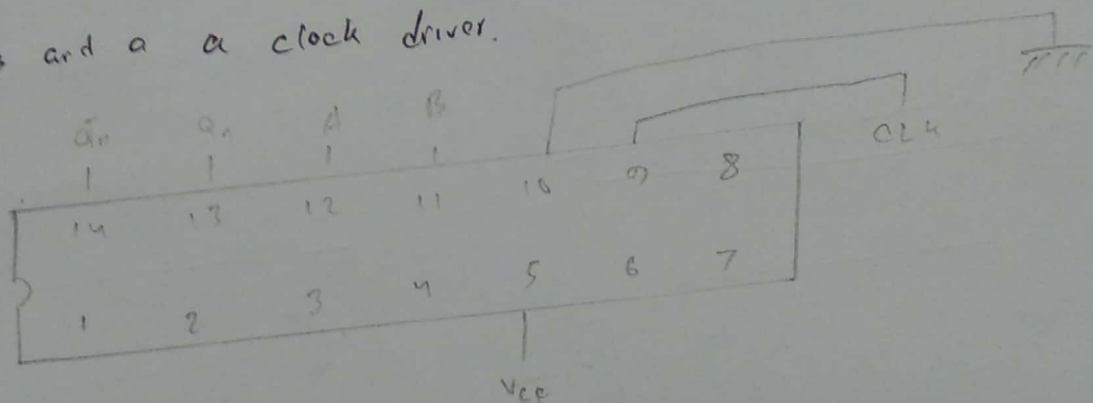


Fig: pin diagram of IC-7491A

This model has maximum clock frequency of 18 MHz and power dissipation of 175 mW. It also has operating free air temperature range of -55°C to 125°C . Single data and input controls are gated through inputs A and B.

and an internal inverter to form the complementary inputs to the first bit of the shift register. Internal common clocking is provided. This clock pulses across the circuit to shift one bit right on every positive edge. The last information of the device is clocked out.

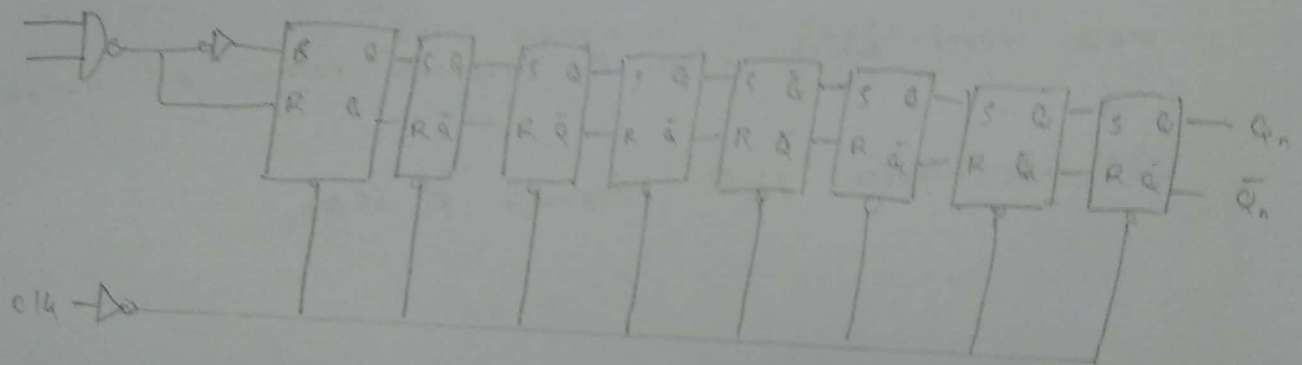


Fig: Logic diagram of IC-7491A

IC-74164:

This is a high speed serial in parallel out shift register. Serial data is entered through a 2 input AND gate where inputs are data (A) and selection bit (B). The input and transition of all ~~data~~ data is synchronized with a positive edge triggered clock pulse. The device has an asynchronous master reset which clears the register setting all output to 0. It has typical shift frequency of 35 MHz.

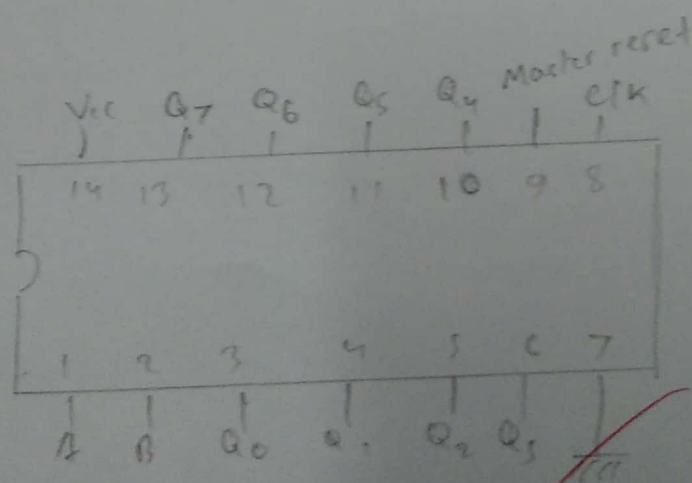


fig: pin diagram of IC-74164

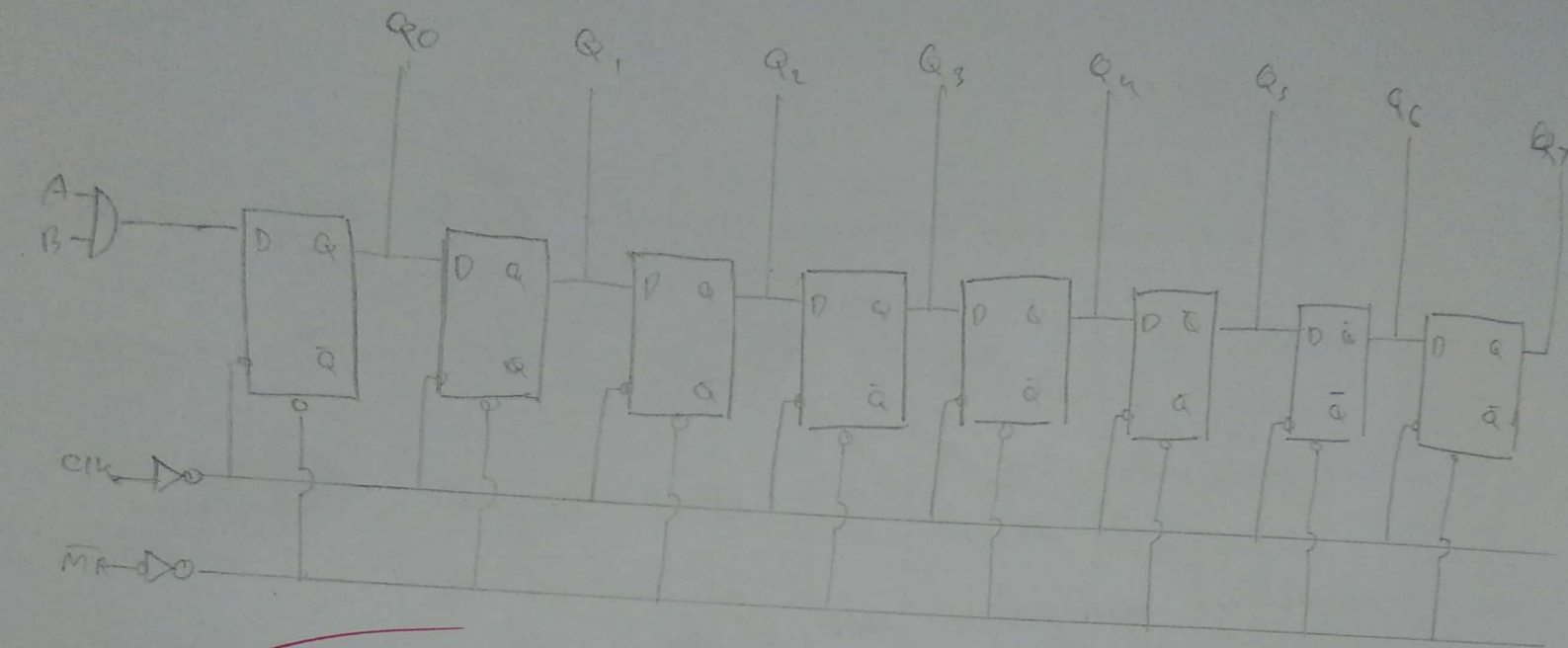


Fig: Logic diagram of IC-74164

Discussion :

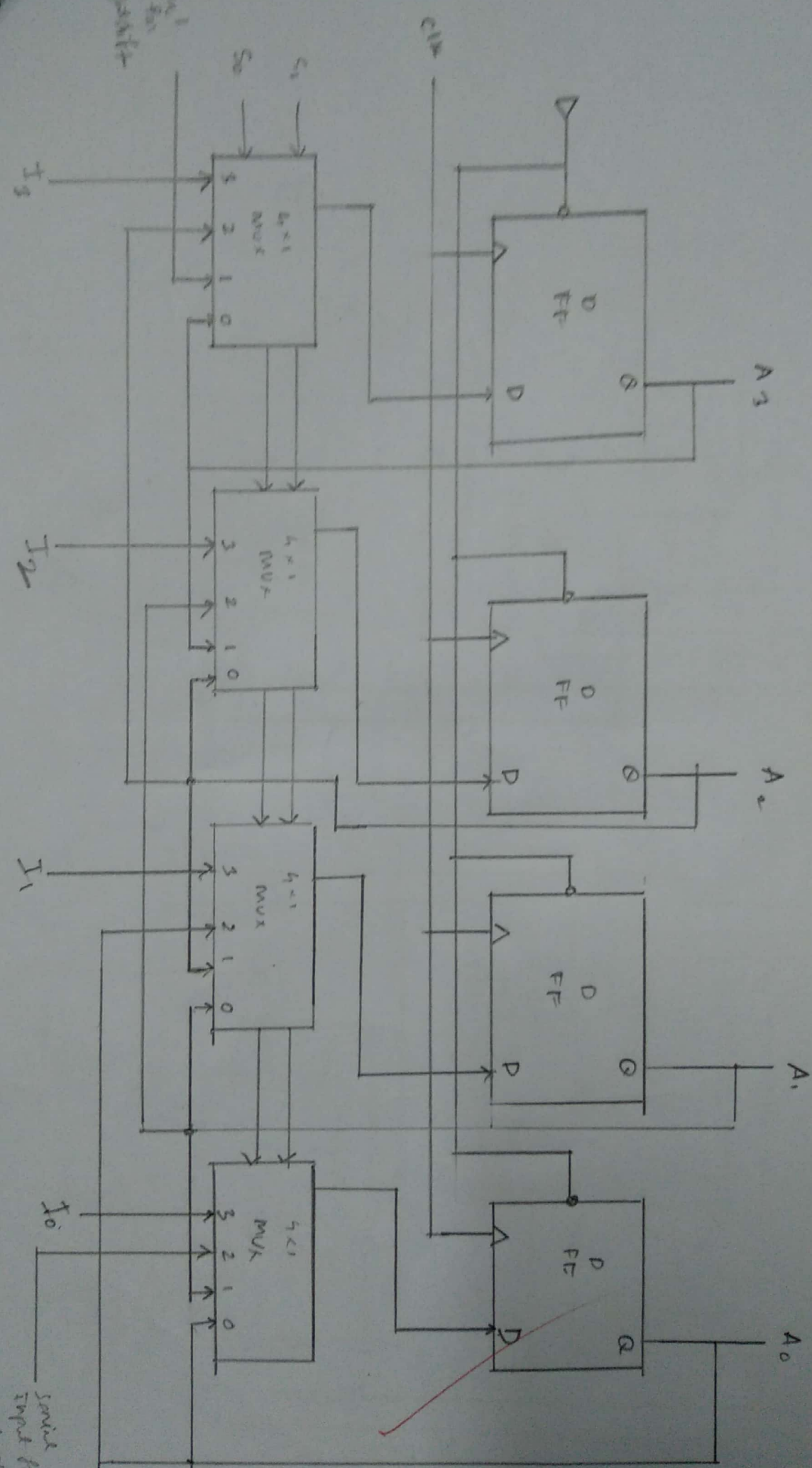
The experiment was implemented according to the circuit design carefully. At first one bit of the 4 bits in the experiment was not working. Then it was solved by making the wire connections more tight. One of the gates was not functioning and that had to be replaced. In conclusion, the experiment was very difficult to implement.

Group: 1

Experiment: 09

INPUT					output				
$\overline{\text{CLEAR}}$	clock	S_0	S_1	Internal signal D_i ($i=0,1,2,3$)	A_3	A_2	A_1	A_0	Mode
0	X	X	X	X	0	0	0	0	Asynchronous CLR
1	X	0	0	X	A_3	A_2	A_1	A_0	data hold
1	$\uparrow$	0	1	A_{i+1}	A_2	A_1	A_0	I_{left}	shift Left
1	$\uparrow$	1	0	A_{i-1}	I_{right}	A_3	A_2	A_1	Shift Right
1	$\uparrow$	1	1	A_i	A_3	A_2	A_1	A_0	Parallel Load

Serial
Input for
Eight



Output

